St Mary's DSG, Kloof

Grade 10 | Physical Science Test
Mole Calculations | 18 May 2022

TIME: 30 min | MARKS: 26

ASSESSOR: K. Bolton | MODERATORS: G. Main and T. Kirsten

NAME:_____

SCIENCE TEACHER:_____

PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 4 pages.
   A data sheet is provided.
   Please check that your question paper is complete.

2. Answer on your answer sheet in the spaces provided.

Question 1

Consider a 201,6 dm^3 sample of carbon dioxide (CO_2) gas at STP. Calculate:

1.1 The number of moles of CO_2. (2)

1.2 The number of molecules present in this sample of CO_2. (2)

1.3 The molar mass of CO_2. (2)

1.4 The mass of CO_2, in this sample. (2)

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Question 2

2.1 The balanced equation for the decomposition of ammonium nitrate is given below:

$$NH_4NO_3(s) \longrightarrow N_2O(g) + 2H_2O(g)$$

2.1.1 In one such decomposition $3{,}01 \times 10^{24}$ molecules of water are released. Calculate the number of moles of water released. (2)

2.1.2 Calculate the number of moles in 240 g of NH_4NO_3. (3)

2.1.3 Calculate the volume of 4 moles of N_2O at STP. (2)

2.2 Ammonia gas is prepared in industry from the reaction of hydrogen gas and nitrogen gas, in the presence of a catalyst, as shown in the equation;

$$N_{2(g)} + 3H_{2(g)} \leftrightarrows 2NH_{3(g)}$$

2.2.1 Calculate the number of moles of hydrogen needed to completely react with 1,5 moles of nitrogen? (2)

2.2.2 What mass of ammonia is formed when 16,8 g of nitrogen reacts with excess hydrogen? (5)

2.2.3 What volume of hydrogen is needed to form 7,84 dm^3 of ammonia at STP? (4)

[18]